Reg. No. | | | | | | | | | | | [MSCCH103]

FIRST SEMESTER M.Sc. (CHEMISTRY) EXAMINATION, DECEMBER 2015

PHYSICAL CHEMISTRY - I

Time : 03 Hours **Maximum Marks:80**

Instructions: 1. Questions **1 to 6** carry **15** marks each. Answer any Four

2. Question No. **7** is compulsory, carries 20 marks.

1. **a.** Explain Nernst heat theorem with its applications and limitations.

b. Discuss the limitations of Van't Hoff's equation. **10+5**

2. **a.** Define and derive Maxwell-Boltzmann statistics.

b. Plot the energy level for various temperatures for Maxwell-Boltzmann, Bose-Einstein and Fermi Dirac statistics.

c. Derive rotational partition function for an ideal gas.

d. Define root mean square velocity with respect to temperature. **5+3+4+3**

3. Explain the following:

a. Surface tension of electrolytic solutions.

b. Effect of high electric field and high frequency on ionic conductance. **7.5+7.5**

4. Explain in detail, the diffusion layer and draw the simplified scheme of the double layer. **15**

5. **a.** Explain the Franck-Condon principle.

b. Write a note on fluorescence and phosphorescence. **10+5**

6. Explain the following:

a. Two dimensional gel electrophoresis.

b. Isolation of proteins on the basis of their isoelectric point.

c. Explain the viscosity of bio molecules and Hofmeister series **7+4+4**

7. **Answer any *FOUR* of the following.**

a. Explain Maxwell relation with respect to:

i. Combined form of I and II law **ii.** Enthalpy and

iii. Helmholtz free energy

b. Discuss briefly the mean free path.

c. Give an account on activity coefficient theory of the electrolytes.

d. Explain the polarization and over voltage.

e. Describe photochemical substitution reactions with suitable examples.

f. Write a note on polyacrylamide gel electrophoresis (PAGE) technique. **5+5+5+5**

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